SORAYA REMAILI

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Education

University of South Carolina Honors College, Columbia, SC
B.S. in Computer Science, B.S. in Statistics, and B.S. in Biology with Honors
Expected May 2027
GPA: 4.0/4.0

Thomas Jefferson High School for Science and Technology, Alexandria, VA
High School Diploma, June 2023
GPA: 4.45 (weighted scale)
AP Scholar with Honor

The Paris Institute for Political Science, Paris, FR
Summer Certification, July 2022
Final Grade: 17.75/20
Focus on International Negotiation and Affairs

Honors and Grants

University of South Carolina President's List, 2023–Present
University of South Carolina Academic Excellence Scholar, $2000/year
University of South Carolina Undergraduate Scholarship, $25000/year
South Carolina Honors College Undergraduate Research Grant, 2024–2026, $6000
Carter Bays Endowed Scholarship for Computer Science, 2024–2025, $1000
Vlahoplus Engineering Honors Scholarship, 2025–2026, $1000
Provost AI Undergraduate Fellowship, 2026, $5000
Magellan Mini Research Grant, 2026, $750
SEAMAMMS Runner-Up Best Undergraduate Long Talk, 2026

Posters and Presentations

Gray's Reef Nat'l Marine Sanctuary Symposium, Richmond Hill, GA **September 2026**
Remaili, S.S.*, A.M. Kreuser, A.A. Sefah-Twerefour, E. Meyer-Gutbrod, and C.R. Edwards.
Update on Real-Time and Archival Acoustic Monitoring of North Atlantic Right Whales Along the Georgia Coast.

International Sclerochronology Conference, St. John's, NL **May 2026**
Wanamaker, A.D., A. Walton, S. Griffin, B. Black, A. Hauser, L. Jarosinski, N.M. Whitney, C.C. Ummenhofer, **S.S. Remaili**, A. Biastoch, T. Martin, and D.L. Thatcher.
Geochemical Signatures in the Long-lived Marine Bivalve *Arctica islandica* Show Intense Warming in the Down East Coastal Region of the Gulf of Maine.

International Sclerochronology Conference, St. John's, NL **May 2026**
Thatcher, D.L., H. Sun, A.D. Wanamaker, N.M. Whitney, L. Jarosinski, J.A. Stewart, B. Williams, M. LaVigne, C.C. Ummenhofer, **S.S. Remaili**, A. Biastoch, and T. Martin.
Master Shell Chronology and Multi-Proxy Geochemical Records Illustrate Oceanographic Variability in the Mid-Atlantic Region (USA) Since 1800 CE.

International Sclerochronology Conference, St. John's, NL **May 2026**
Keynote Presentation
Whitney, N.M., A.D. Wanamaker, D.L. Thatcher, B. Williams, C.C. Ummenhofer, **S.S. Remaili**, A. Biastoch, T. Martin, H. Sun, L. Jarosinski, J. Berg, A. Walton, M. LaVigne, J. Stewart, E. Powell, R. Mann, E. Hofmann, J. Klinck, N. Sprague, M. Spencer, A.K. Watkins, B. Martin, A. Hauser, R. Copich, and B. DePhillips.
Utilizing a Network of *Arctica islandica* Records and Model Simulations to Investigate Past Re-

gional Ocean Dynamics in the Rapidly Warming Western North Atlantic.

SEAMAMMS, Wilmington, NC **April 2026**
Oral Presentation
Remaili, S.S.*, A.M. Kreuser, A.A. Sefah-Twerefour, E. Meyer-Gutbrod, and C.R. Edwards.
Lessons from Real-Time and Archival Acoustic Detections of North Atlantic Right Whales in the Southeast US across Three Types of Platforms.

Discover USC, Columbia, SC **April 2026**
Remaili, S.S.*, A.M. Kreuser, A.A. Sefah-Twerefour, E. Meyer-Gutbrod, and C.R. Edwards.
Lessons from Real-Time and Archival Acoustic Detections of North Atlantic Right Whales in the Southeast US across Three Types of Platforms.

American Geophysical Union, New Orleans, LA **December 2025**
Remaili, S.S.*, N.M. Whitney, B. Williams, C.C. Ummenhofer, A. Biastoch, T. Martin, A.D. Wanamaker, D.L. Thatcher, J. Berg, A. Hauser, H. Sun, A. Walton, M.J. Mette, E. Powell, R. Mann, E. Hofmann, and J. Klinck.
Explaining Variability in Shell-Based Oxygen Isotope Ratios in the Western North Atlantic Using a Proxy System Model.

Discover USC, Columbia, SC **April 2025**
Remaili, S.S.*, A.M. Kreuser, A.A. Sefah-Twerefour, K. Dreger, E. Meyer-Gutbrod, and C.R. Edwards.
Pilot Program for Glider-Based Passive Acoustic Monitoring of Right Whales in the Southeast US Calving Ground (2025 version).

North Atlantic Right Whale Consortium, Providence, RI **October 2024**
Meyer-Gutbrod, E., C.R. Edwards, A.M. Kreuser, A.A. Sefah-Twerefour, K. Dreger, and **S.S. Remaili**.
Pilot Program for Glider-Based Passive Acoustic Monitoring of Right Whales in the Southeast US Calving Ground (2024 version).

Research Experience

Active Acoustic ID of Arctic Fauna **May 2026–Present**
Fulbright Canada-Mitacs Internship at Memorial University of Newfoundland
Research Directors: Dr. Frédéric Cyr and Dr. Maxime Geoffroy
- Analyze broadband acoustic profiles to identify fish and zooplankton across the polar front.
- Use Echoview and Python to process acoustic data from 38 kHz and 333 kHz transducers.
- Compare acoustic size estimates with spectra from LISST, LOPC, and UVP instruments.

Skills: Echoview, broadband acoustics, active acoustic data processing, unsupervised classification, CTD-Rosette/WBAT data.

Resilient Water Futures Lab **February 2026–Present**
USC School of the Earth, Ocean, and Environment
Research Director: Dr. David Fuente
- Model Nairobi water billing and payment behavior using customer records from 2012–2023.
- Build Gaussian mixture models to classify customers into interpretable payment typologies.
- Identify high-arrears and low-payment customer groups to support utility revenue strategies.

Skills: XGBoost regression trees, Gaussian mixture models, customer segmentation, feature engineering, Python ML libraries.

Paleoceanographic Modeling **June 2025–August 2025**
NSF-REU at Shannon Point Marine Center
Research Director: Dr. Nina Whitney

- Used Python to model paleoceanographic data and analyze historical climate variability.
- Applied multi-taper spectral analysis to identify patterns in climate and oceanographic records.
- Presented research findings at scientific symposiums and strengthened scientific communication skills.

Skills: Climate modeling, Python, Pyleoclim package usage, climate data product wrangling, multi-taper spectral analysis.

NOAA Real-Time Analysis Workshop **May 2024**

- Gained hands-on experience with real-time data analysis techniques for environmental monitoring.
- Collaborated with NOAA scientists and peers to analyze marine ecosystem data through simulations and case studies.

Conservation Oceanography Lab **March 2024–Present**

USC School of the Earth, Ocean, and Environment
Research Director: Dr. Erin Meyer-Gutbrod

- Analyze acoustic data to identify baleen whale presence using Python, Raven, R, and LFDCS.
- Compare detection methods by accuracy to evaluate their usefulness for whale conservation research.
- Process and classify real-time acoustic data while developing statistical analysis and visualization skills.

Skills: Low-Frequency Detection Classification Software, R, RStudio, Python, Raven, real-time analysis, acoustic data processing, UNIX/LINUX.

Community Experience

STAT 201: Elementary Statistics Lab TA **Fall 2026**

UofSC Statistics Department

- Develop study guides and exam-review materials to help non-major statistics students prepare for assessments.
- Supported lab instructors by assisting with experiments and answering student questions.

Service Chair/DEI Chair **Spring 2024/Fall 2024**

Theta Tau Professional Engineering Fraternity

- Co-led Theta Tau's service and equity committees, aligning projects with organizational goals.
- Managed budgets and resources for service projects, securing additional funding when needed.

Technology Adventure Camp Assistant Instructor **July 2024**

Fairfax County Public Schools Summer Programs

- Taught children foundational robotics concepts, coding, and sensor technology in a hands-on environment.
- Designed and facilitated engaging activities to teach computer skills and programming languages.

Science Outreach

Anacortes & Oak Harbor Public Library Touch Tank **July 2025**

- Designed and executed a marine habitat learning experience for 100+ children.

Guardianes Del Mar STEM Program **June 2025**

- Helped facilitate an intertidal zone and microscopy workshop for 15 middle-school students.

North Atlantic Right Whale Festival **November 2024**

- Designed and taught activities to 300+ visitors to teach the importance of bioacoustics.

Projects

Pathways: Campus Connection Made Easy

- Built a full-stack student platform integrating campus navigation, events, courses, calendars, and career planning tools.
- Developed the system using React, Django REST Framework, Docker, Redis, Celery, and

GitHub Actions for deployment and automation.

DegreeBetter Academic Management System

- Full, back-to-frontend development of a degree planning system using Java, FXML, and Git.
- Developed the backend system using Java to handle user data, course requirements, and degree tracking.

Personal Website and Portfolio

- Built and deployed a responsive portfolio website using HTML, CSS, and JavaScript.
- Used GitHub Pages for hosting and Git for version control.

Skills

French: Professional working proficiency; Virginia State Seal of Biliteracy Recipient
Algerian Darija: Limited working proficiency; completed courses at Johns Hopkins CTY
Technical: R/RStudio, Python, C++, Java/JavaFX, SQL, Git & GitHub, Raven, HTML/CSS, FXML, LFDCS, Echoview, MATLAB
CITI & USC Certifications: Physical Science Responsible Conduct of Research; Social and Behavioral Responsible Conduct of Research; Garnet AI Fluency